

Sharanya Nagar

Full-Stack Developer & Software Engineer

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SUMMARY

Full-stack software engineer with hands-on experience building distributed, concurrent systems and AI-integrated applications. Designed and deployed production-grade infrastructure for scalable, secure platforms using Python, Java, and JavaScript. Passionate about large-scale systems, applied ML, and building software that operates at the intersection of reliability and performance.

EDUCATION

Bachelors in Technology CSE(AI-ML)

Adani University
9.20 CGPA

2024 – Present

Diploma in Programming

Indian Institute of Technology, Madras
Dual degree

2024 – Present
Online

SKILLS

Languages — Python • Java • JavaScript • SQL • C++

Frameworks and Libraries — React.js • Flask • Node.js • Express.js • FastAPI • scikit-learn • PyTorch

Databases and Infrastructure — PostgreSQL • Redis • Celery

Standards and Security — CSRF Protection • Rate Limiting • RBAC • REST APIs

CS Fundamentals — Data Structures • Algorithms • System Design • Concurrency • Async Programming

PROJECTS

AION Code Reviewer

Tech: Python • PyTorch • Gymnasium • GPT-2 • AST • GitHub Actions

- <https://github.com/Wall-E-30/code-reviewer> 
- Designed a custom Gymnasium-compatible environment to train LLMs on autonomous code review via a 2-phase pipeline (Supervised Fine-Tuning + REINFORCE with variance-reducing baseline), achieving 0.98+ cumulative reward across 3 benchmark tasks.
- Built a static safety sandbox with AST-based grading and regression backtracking, blocking 15+ unsafe execution patterns across 13 programming languages.
- Validated pipeline stability with 18 automated integration tests executed on every commit via GitHub Actions, covering reward signals, AST grading edge cases, and regression recovery.
- Medium Blog: <https://medium.com/@sharanyanagar/we-built-aion-an-ai-that-teaches-itself-to-fix-your-code-using-reinforcement-learning-57ae8564518f> 

Arogya: Hospital Management System

Tech: Flask • React • Socket.io • Postgres • Docker • CI/CD pipeline

- <https://github.com/sharancreates/hospital-management-system> 
- Architected a distributed, concurrent backend using Flask + Celery + Redis, enabling asynchronous background job processing across role-based workflows for patients, doctors, admins, and nurses — designed to handle high-throughput scheduling and billing operations at scale.
- Implemented standard healthcare interoperability protocols (HL7 v2, FHIR R4) for EHR compatibility — enabling clinical data exchange used in production hospital systems globally.
- Medium Blog: <https://medium.com/@sharanyanagar/engineering-arogya-lessons-from-building-a-secure-healthcare-platform-ced0cf4ccca2> 

ResuMatch AI

Tech: Python • FastAPI • spaCy • sentence-transformers • PostgreSQL

- <https://github.com/sharancreates/resumatch> 
- Engineered a hybrid ATS matching engine blending lexical n-gram tokenization (40%) with transformer embeddings (60%), improving resume-to-job relevance by 35% over keyword-only matching, with a thread-safe LRU embedding cache cutting scan latency by 60% (~250ms per scan).
- Boosted entity extraction accuracy to 95%+ by building a custom spaCy NLP pipeline with a PhraseMatcher covering 100+ technical skill patterns, replacing fragile regex-based parsing.

Selene: A Menstrual Cycle Tracking Application

Tech: Python • scikit-learn • PostgreSQL • Redis • Flask • GitHub Actions

- <https://github.com/sharancreates/selene> 
- Improved cycle forecasting accuracy by training a Random Forest regressor on historical symptom vectors, reducing prediction error to ± 1.5 days — a 45% improvement over standard static 28-day tracking heuristics.
- Implemented multivariate anomaly detection using an Isolation Forest model to flag extreme symptom spikes and logging gaps, providing early-warning telemetry for chronic conditions (PMOS, PMDD, Endometriosis).
- Achieved zero-knowledge data storage by encrypting 100% of health records before database serialization, using client-side key derivation stretched via 100,000 PBKDF2-HMAC-SHA256 iterations to resist brute-force attacks.
- Reduced API response latency by ~30% through PostgreSQL connection pooling (size=10, overflow=20) and targeted indexing on high-frequency query paths (user_id, log_date).

CERTIFICATES

Cloud Computing

IBM SkillsBuild

POSITION OF RESPONSIBILITY

Web Development Team Member in Tech Club

Mentored junior members in core web technologies (HTML, CSS, JS), conducting code reviews and guiding them through their first project deployments. Accelerated personal proficiency in the MERN stack by architecting scalable solutions for club events, translating theoretical knowledge into production-grade code

EXTRA CURRICULAR ACTIVITIES

Social Impact

Explored and conceptualized digital solutions for "Peace, Justice, and Strong Institutions" (SDG 16), focusing on legal awareness and civic engagement platforms like Digital Sakshya.

Creative Writing

Actively write and publish poetry and prose, honing the ability to articulate complex abstract concepts into clear, compelling narratives

OPEN SOURCE CONTRIBUTIONS

Submitted PR to `condor_code` — Refactored UI components for test screens and improve web compatibility | https://github.com/CondorCodeOrg/condor_code 

HACKATHONS

AETRIX 2026

Tech: FastAPI | GEE API | React | PostGIS | PostgreSQL | Redis | Celery | scikit-learn

Reached the final round | Built *Pratyaksha*— A B2G application which interprets satellite data, and suggests the actions to be taken for environment using ML models.

TarkShaastra 2026

Tech: FastAPI | React | PostgreSQL | Redis | Celery | scikit-learn

Participant | Built *Complaint Management System* which auto allocates complaints in consumer care and solves them by past data using ML models, supervised by a human.